

MAE 1001



Introduction to Mechanical and Aerospace Engineering

Course website: <https://gwu-mae1001.github.io/fall2026/>

Prof. Kartik Bulusu, MAE Dept.

Graduate Teaching Assistant:

Preethi Siva Kumar (MAE Dept., MS student)

Learning Assistants:

Anya Spevacek (Senior, MAE Dept.)

Ergi Agaj (Senior, MAE Dept.)

Toby Pike (Junior, MAE Dept.)



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Fall 2026

Photo: Kartik Bulusu

Road Map for MAE 1001

Photo Credits: Kartik Bulusu

Icon source: thenounproject.com

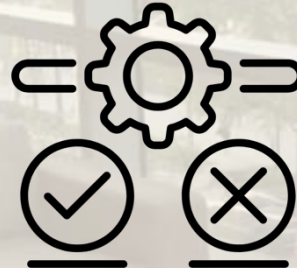
Icon credits: LUTFI GANI AL, Studio 365, DesignBite, Danil Polshin, Sofia



Exposure to various career paths in
Mechanical Engineering



Teamwork and
Problem solving



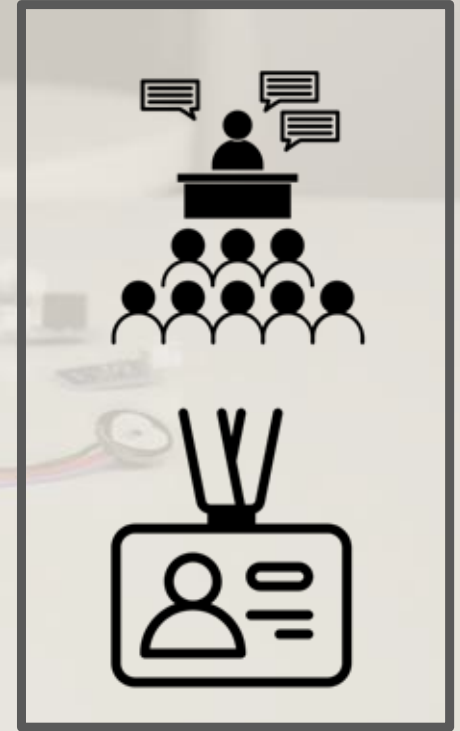
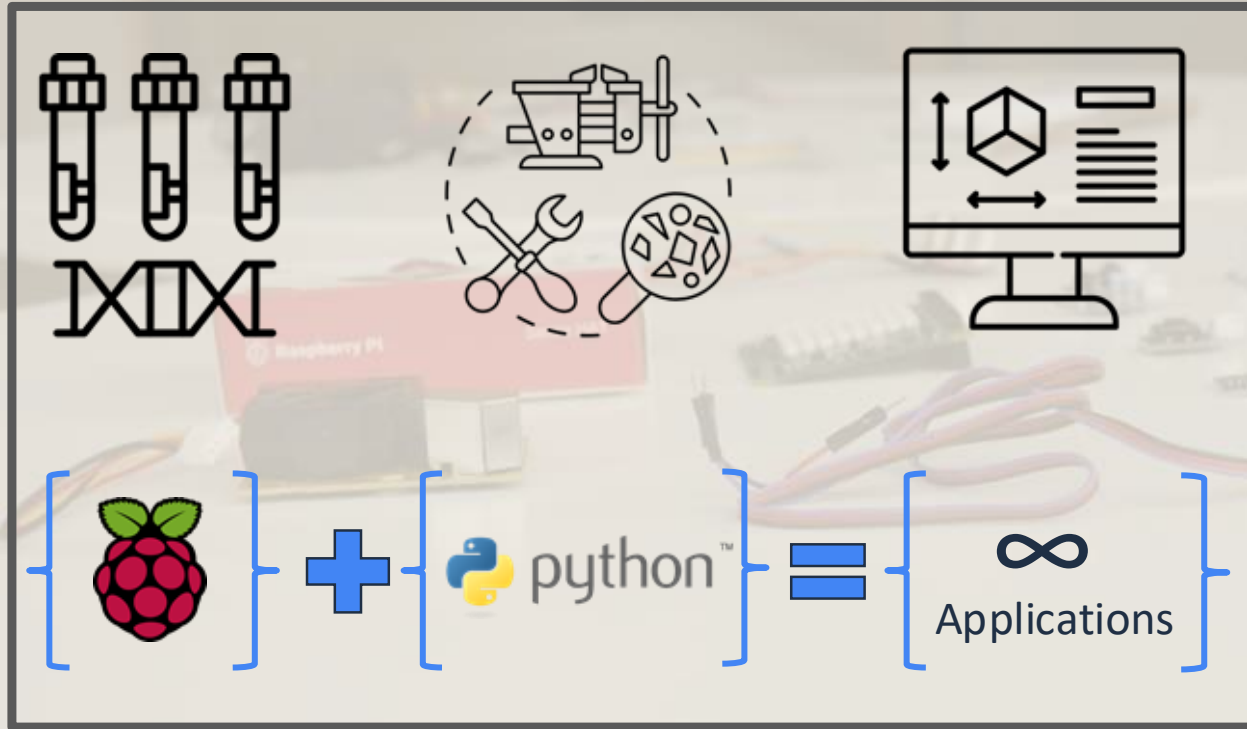
Awareness of
Engineering Ethics



Computational thinking,
Hardware and
Programming skills



What's in our 8-week journey?



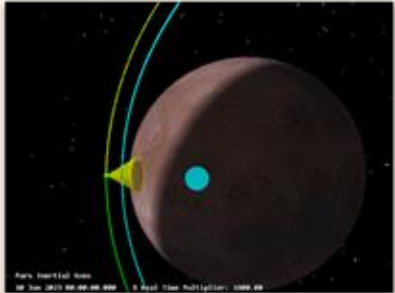
MAE 1001 Instruction Team



MAE 1001 Instruction Team



Biofluid
Dynamics
Laboratory



accenture



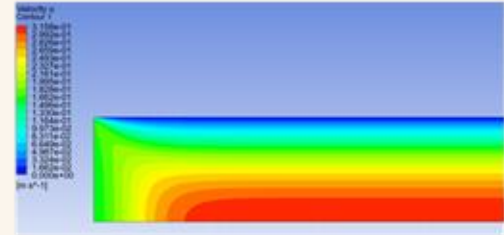
GE HealthCare



30 Countries



10 Hours



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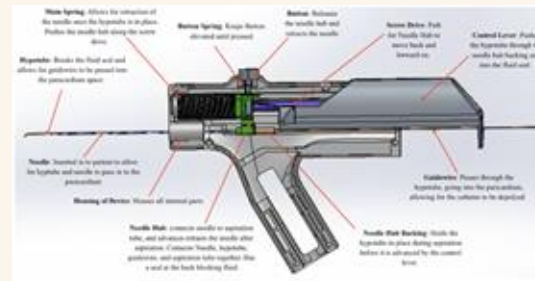
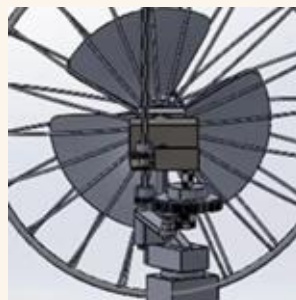
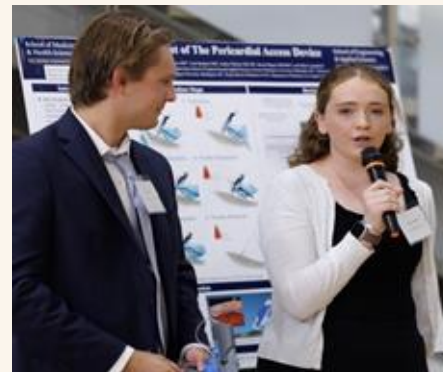
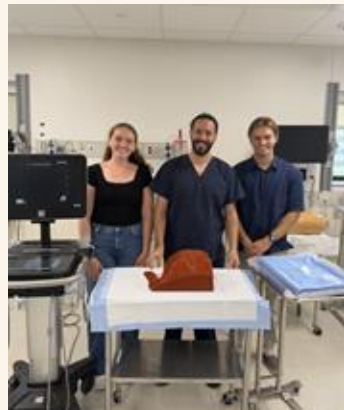


Prof. Kartik Bulusu, MAE Dept.

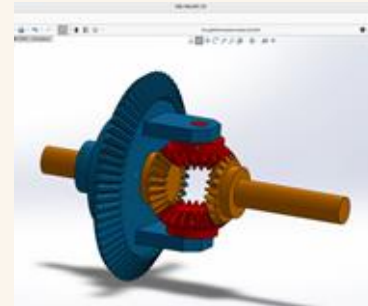
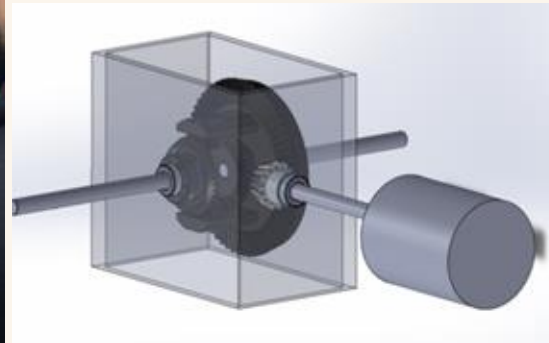
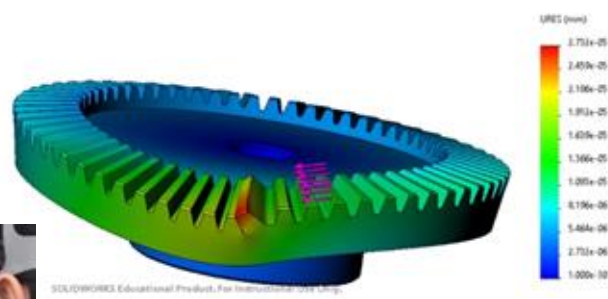
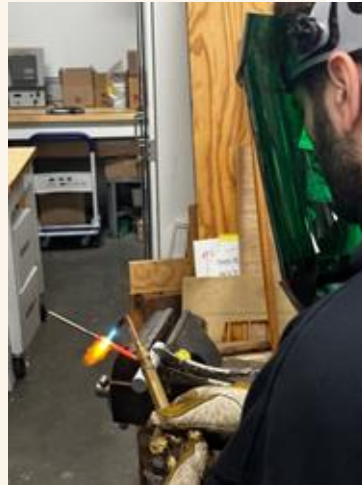
Fall 2026

MAE 1001 Introduction to Mechanical and Aerospace Engineering

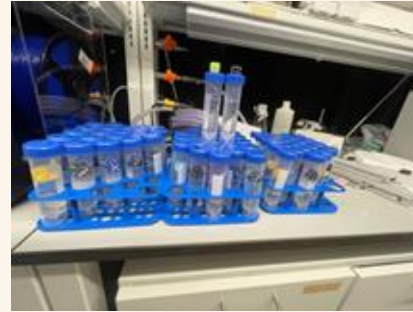
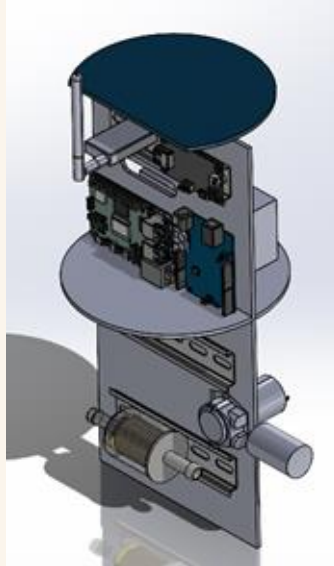
Instruction Team: Anya Spevacek



Instruction Team: Ergi Agaj



Instruction Team: Toby Pike



Instructor is like an Object in a Python program

- Everything in Python is an **object**
 - **number 9**, is an object
 - **range()** is an object
- **Objects** can be manipulated
- **Objects:**
 - have a **type()**, that determine what python program can do with them

type



human

Analogy

type



hobbit

- Python has two kinds of objects
 - **scalar**: very basic and other objects can be made with them
 - **non-scalar**: have an internal structure



Ground Rules

How we work together this semester

Be Respectful

- Listen to the instructors
- Keep an open mind
- Limit the use of personal devices

Be Responsible

- Arrive on time
- Follow through with instructions
- Help each other

Be a Communicator

- Observe, ask and try
- Let us know of emergencies
- Use EdStem and discuss

Be a Problem Solver

- Explore options
- Make your own notes
- Stay positive and have fun



MAE 1001 · WEEK 0

Announcements

A few opportunities, friendly faces, and next steps to help you settle in.

01

**Ethics
Competition**

02

Guest Speakers

03

Spring Advising

04

Office Hours



GWU IEEE × IEEE SSIT · STUDENT ETHICS COMPETITION

NO EASY ANSWER. DEFEND YOURS.

One real engineering ethics case. One hour to crack it with your team. Ten minutes to convince a panel of IEEE leaders, then five more to survive their questions.

\$50

per teammate for 1st place,
plus certificate

8

team spots only, first come,
first served

\$0

to enter, and the pizza
lunch is on us

SIGN UP RIGHT NOW

SCAN BY SUN, OCT 11



3-minute interest form. Zero commitment.
No team? Sign up solo and we'll match you.

Guest speaker: Zain Mohsin



Senior

Department: Mechanical and Aerospace Engineering

Email: zain.mohsin@gwu.edu

LinkedIn: [linkedin.com/in/zain-mohsin-a363472aa/](https://www.linkedin.com/in/zain-mohsin-a363472aa/)

Hi, my name is Zain Mohsin and I am a Senior in Mechanical Engineering from Michigan. I love being creative, being active, and working on designs for the GW Design/Build/Fly team.





GW Design/Build/Fly

Jack Barbano (Team Lead)

Zain Mohsin (Chief Engineer & Capstone Team Lead)



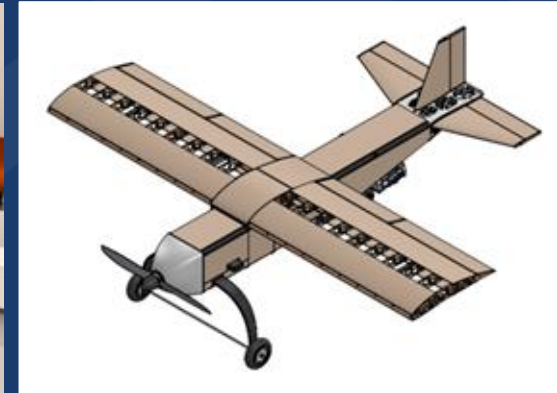
What does DBF do?

- Each year, we design, build, and fly a new remote-controlled UAV to compete in the annual AIAA Design/Build/Fly Competition.
- Held in the Spring in Tucson, AZ / Wichita, KS.
- New requirements each year with new missions that requires different systems
 - e.g. autonomous glider, deployable banner, etc.



GW DBF 2025-26

- Passengers and cargo
 - Rubber ducks and hockey pucks
- Deployable banner
 - Release while in flight and then drop



GW DBF 2026-27

- Towed sensor
 - Module with lights deployed in flight
- Shipping container
 - Protective container for sensor
 - Transported inside plane



Early conceptual CAD

Why DBF?

- Gain practical skills
 - Hands-on build project/technical experience
 - In-depth design and analysis
 - Collaboration and leadership
- Internships and professional life
 - Aerospace: commercial aircraft, UAVs, eVTOLs, experimental aircraft
 - Everything else!
- Have fun with your work
 - Build and fly planes, go to competitions





GW DBF Interest Form



GW AIAA Interest Form

Guest speaker: Professor Megan Leftwich



Professor

Department: Mechanical and Aerospace Engineering

Email: mleftwich@gwu.edu

Office Phone: (202) 994-9830

Location: SEH 3880 | Office hours: Wednesday: 1:00 - 3:00 pm

Professor Megan Leftwich's Biologically Inspired Energy Laboratory investigates natural fluid dynamics phenomena and applies their findings to engineering problems. Current projects include the wake dynamics of vertical axis wind turbines in standard and complex configurations, the hydrodynamics of pinniped swimming (in collaboration with the Smithsonian National Zoo), and the fluid dynamics of human birth.



Guest speaker: Jonathan Yoo



Senior Academic Advisor

Department: Undergraduate Academic Advising and Student Success

Email: seasadvising@gwu.edu

Office Phone: (202) 994-6158

Location: Science and Engineering Hall Suite 2500

Jonathan received his Bachelor of Arts in International Studies at Bowling Green State University and Master of Arts in International Education at The George Washington University. He has over 6 years of work experience in the higher education field, focusing on academic advising and intercultural training.



Second Semester Schedule

- MAE 1004 - Engineering Drawing and Computer Graphics (3 credits)
- MAE 1117 - Introduction to Engineering Computations (3 credits)
- MAE 1125 - Applied Linear Algebra for Engineering (3 credits)
- PHYS 1021 - University Physics I (4 credits)
- Non-Technical Elective - Humanities, Social Science, or additional (3 credits)

Second Semester Schedule - Math

- If you are in MATH 1220, you need to take MATH 1221 (with MATH 1232 in the summer)
- If you are in MATH 1231, you need to take MATH 1232 (OPTIONAL – MATH 2233 in the summer)
- If you are in MATH 1232, you need to take MATH 2233
- If you are in MATH 2233, you need to take APSC 2113 (OPTIONAL)

****Option to take at GW or at another institution as transfer credits***

Spring 2027 Registration Dates

- Your registration date will be determined by the amount of credits you have earned (not in-progress). Here are the dates:
 - November 6 (Friday) - Advanced Registration
 - November 9 (Monday) - 90 or more credits earned
 - November 10 (Tuesday) - 70 or more credits earned
 - November 11 (Wednesday) - 50 or more credits earned
 - November 12 (Thursday) - 30 or more credits earned
 - November 13 (Friday) - 0 or more credits earned
 - November 16 (Monday) - General Registration Begins, Waitlist Open, First day to submit RTF-EZ

Friendly Reminders

- GW Schedule of Classes for Spring 2027 is available to view
- Review your DegreeMAP very carefully
- Create 2-3 Plan Ahead schedules for Spring 2027
- Complete and submit your Undergraduate Advising Form (UAF) for Spring 2027 – make sure to complete correctly
 - Must be submitted and approved BEFORE your registration date
 - Failure to submit your UAF by Sunday, November 22 will result in a HOLD on your account
- Registration opens at 9:00 AM on your assigned date.

SEAS ADVISING · WE ARE HERE FOR YOU

Thank you! Questions?

Email questions to the GW Engineering Undergraduate Advising Team at seasadvising@gwu.edu.

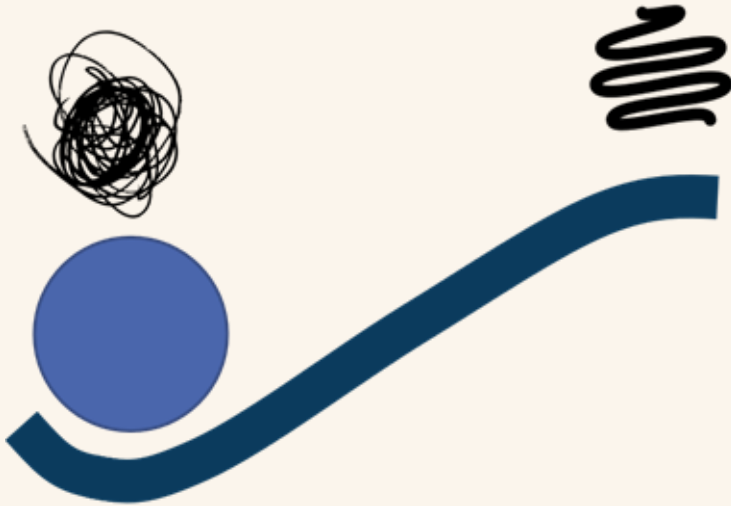
Schedule a 1:1 Meeting – <https://calendly.com/jyoo7>

Drop-In Advising Hours: Wednesdays from 2:00-4:00 PM

Office location: Science & Engineering Hall, Suite 2900
(2nd floor on the right side of the elevators)

Join EdStem

edstem.org/us/join/qFPHsX



MAE 1001 • FALL 2026

Office Hours

Questions big or small — come say hi.
We'd love to see you.

MONDAY

3:30 – 4:30 PM

Anya Spevacek

Undergraduate LA

TUESDAY

11 AM – Noon

Ergi Agaj

Undergraduate LA

TUESDAY

2:30 – 3:30 PM

Toby

Undergraduate LA

THURSDAY

1:00 – 2:00 PM

**Preethi Siva
Kumar**

Graduate TA



Prof. Kartik Bulusu

After class, or by appointment — just reach out.
bulusu@email.gwu.edu

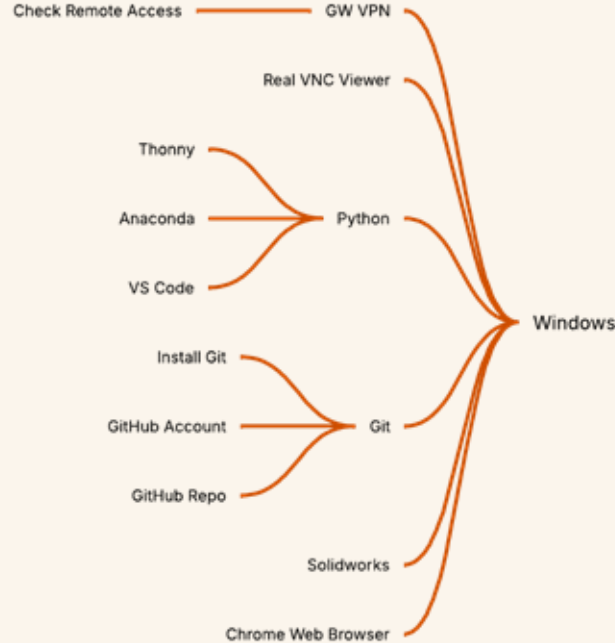


SEH, 3rd floor

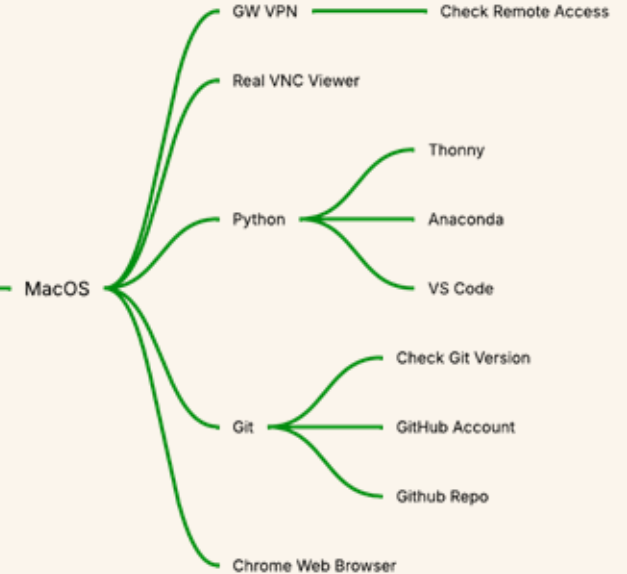
By the couches next to SEH 3040.
Can't make it? Message us on **EdStem** or
Blackboard.



Software installation clinic



Software Installations



Be a part of a Raffle drawing



What should you expect in the next 8 weeks?

Kick off exercises

Exposure {
 To Student life
 To Departmental Life
 To Departmental Research

Skills {
 To co-work with colleagues
 To Make
 To Implement

Transformational experience

What your instructor expects in the next 8 weeks?

Observe, Ask and Try

Identify {
 Challenges {
 Opportunities {
 Gaps {
 Surprises {
 Obvious
 Hidden
 Entrepreneurial
 etc.

Build cool hacks with Raspberry Pi

Accomplish tasks with Python Programming

What did you learn today? A lingering question?

